P-10
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Code No: 227AC

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Supplementary Examinations, May-2026

DATA MINING

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours | Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

	PART- A	**(20 Marks)**
1. a)	Define Data Mining.	[2M]L1
b)	What is data transformation?	[2M]L1
c)	Define frequent itemset.	[2M]L1
d)	What is constraint-based mining?	[2M]L2
e)	Define Rule-Based Classification	[2M]L1
f)	What is Rule Pruning?	[2M]L1
g)	What is K-Means Clustering?	[2M]L1
h)	Differentiate Hierarchical and Density-Based Clustering.	[2M]L2
i)	Define Time-Series Data.	[2M]L1
j)	What is Web Usage Mining?	[2M]L2
	PART- B	**(5 X 10 =50 Marks)**
2. a)	Explain integration of Data Mining systems with Data Warehouses.	[5M]L2
b)	Apply data integration techniques for combining multiple data sources.	[5M]L3
	OR	
3. a)	Analyze different issues related to Data Mining systems.	[5M]L4
b)	Evaluate the role of Interestingness Patterns in Data Mining.	[5M]L5
4. a)	Explain Mining Methods used in Association Rule Mining.	[5M]L2
b)	Apply correlation analysis techniques for transactional databases.	[5M]L3
	OR	
5. a)	Analyze various kinds of association rules with suitable examples	[5M]L4
b)	Evaluate Constraint-Based Association Mining techniques	[5M]L5

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6. a) Explain Bayes Theorem with suitable example [5M]L2

b) Apply Naïve Bayes Classification for sample datasets. [5M]L3

OR

7. a) Analyze Lazy Learner methods in classification. [5M]L4

b) Evaluate Rule Pruning techniques in classification problems. [5M]L5

8. a) Explain Categorization of Major Clustering Methods. [5M]L2

b) Apply Density-Based Methods for clustering sample datasets. [5M]L3

OR

9. a) Analyze Partitioning Methods with suitable examples [5M]L4

b) Evaluate Hierarchical clustering techniques for large datasets. [5M]L5

10. a) Explain Mining Time-Series Data with suitable applications [5M]L2

b) Apply Web Content Mining techniques for web applications. [5M]L3

OR

11. a) Analyze Mining the World Wide Web techniques with suitable examples. [5M]L4

b) Evaluate Multimedia Data Mining applications in real-world systems [5M]L5

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Supplementary Examinations, May/June-2026

DATA MINING

(COMPUTER SCIENCE AND ENGINEERING)

Max. Marks: 75

Time: 3 hours

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1. a)	Define Data Mining.	[2M]L1
b)	List and elaborate the challenges in Data Mining.	[3M]L2
c)	How do you calculate support and confidence in association rule mining.	[2M]L1
d)	What is Constraint-Based Mining?	[3M]L1
e)	Compare prediction and classification with example	[2M]L2
f)	What is Naïve Bayes assumption?	[3M]L1
g)	Identify the situations where K-means clustering algorithm is not suitable.	[2M]L3
h)	What is an Outlier?	[3M]L1
i)	What kinds of association can be mined in multimedia data?	[2M]L1
j)	Define web content mining.	[3M]L1

PART- B

(5 X 10 =50 Marks)

2. a)	Describe Data Mining along with its functionalities.	[5M]L1
b)	Explain different Data Preprocessing techniques with examples.	[5M]L2
	OR	
3. a)	What is interestingness in data mining? Explain about it.	[5M]L1
b)	How do data mining and data warehousing work together?	[5M]L1
4. a)	Explain the FP-Growth Algorithm using a suitable sample dataset?	[5M]L2
b)	Can we design a method that mines the complete set of frequent item sets without candidate generation?	[5M]L4
	OR	
5. a)	Compare Association Rule Mining with Classification Techniques.	[5M]L2
b)	What is graph pattern in data mining?	[5M]L1

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6. a) How to evaluate classification algorithm? [5M]L1
b) Describe the data classification process with a neat diagram. [5M]L3

OR

7. a) Classify COVID-19 dataset into +ve(1) or –ve(0) with attributes {fever, body pain, breathing trouble, dry cough, loss of taste or smell} using Suitable classification algorithm. [5M]L3
b) How does the Naive Bayes classification works? Explain. [5M]L1

8. a) Explain how cluster analysis helps in preprocessing data for association mining. [5M]L4
b) What are the disadvantages of agglomerative hierarchical clustering algorithm? [5M]L1

OR

9. a) Describe how categorization of major clustering methods is done [5M]L2
b) Define outlier? Identify outliers in the temperature dataset {21°C, 28 °C,-4 °C, 32 °C, 5 °C,34 °C } [5M]L1

10. a) What are the 4 components of time series data? [5M]L1
b) Discuss the following a. Web structure mining. b. Web usage mining. [5M]L4

OR

11. a) Examine the process for Mining sequence patterns in Transactional databases. [5M]L4
b) Illustrate about following terms. a. Spatial data mining b. Mining world wide web. [5M]L2

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech III Year II Semester Supplementary Examinations, December-2025/January-2026.

DATA MINING

(Information Technology)

Time: 3 hours | Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(10×2 = 20 Marks)

1. a) Define Data Mining. [2M] L1
   b) List out the applications of data mining. [2M] L1
   c) Define support and confidence in Association rule mining. [2M] L1
   d) What are the strategies for data transformations? [2M] L1
   e) Write the difference between classification and prediction. [2M] L1
   f) Write the advantages of decision tree. [2M] L1
   g) What do you mean by cluster analysis? [2M] L1
   h) What is an outlier? [2M] L1
   i) Discuss the primary components of a time series. [2M] L1
   j) What is trend in data mining? [2M] L1

PART- B

(5 X 10 =50 Marks)

2. a) Describe the Functionalities in Data Mining. [5M] L2
   b) Explain in detail about Data Pre-processing. [5M] L3

OR

3. a) Explain how to Integration of Data mining system with a Data warehouse. [5M] L2
   b) Describe in detail the major issues in data mining. [5M] L2

4. Apply apriori algorithm to find frequent item sets using candidate generation for the following data Assume the minimum support count is 2. [10M] L4

TID	List os Item ID's
T100	I1,I2,I5
T200	I2,I4
T300	I2,I3
T400	I1,I2,I4
T500	I1,I3
T600	I2,I3
T700	I1,I3
T800	I1,I2,I3,I5
T900	I1,I1,I3

OR

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5.	a)	Explain about Frequent Patterns mining.	[5M]	L3
	b)	Describe constraint-based Association mining.	[5M]	L2
6.	a)	Discuss Bayesian classification method with an example.	[5M]	L2
	b)	Discuss the process of measuring the accuracy of a classifier.	[5M]	L2
		OR		
7.	a)	Illustrate decision tree induction algorithm with an example.	[5M]	L3
	b)	Explain what is a hyperplane in the context of SVM?	[5M]	L3
8.	a)	Illustrate about different types of clusters.	[5M]	L1
	b)	Illustrate the basic agglomerative hierarchical clustering algorithm.	[5M]	L1
		OR		
9.		Difference between K-means and K-mediods clustering algorithm with example	[10M]	L1
10.	a)	Describe Spatial Data mining with an example.	[5M]	L1
	b)	Explain about Multimedia Data mining.	[5M]	L1
		OR		
11.	a)	Describe about mining time-series data.	[5M]	L1
	b)	What is a moving average, and how is it used in time series analysis?	[5M]	L1

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year II Semester Regular Examinations, April / May -2024

DATA MINING

(INFORMATION TECHNOLOGY)

Time: 3 hours Max. Marks: 75

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A

(25 Marks)

1. a) What is Data mining? [2M] L1

   b) How is a data warehouse different from a database? How are they similar? [3M] L1

   c) Explain support and confidence in Association rule mining. [2M] L2

   d) How to compute confidence for an association rule X -> Y? [3M] L1

   e) Write about nodes in a tree. [2M] L1

   f) Summarize the issues of decision tree learning. [3M] L2

   g) Write about k-Medoids clustering algorithm. [2M] L1

   h) List the different types of data used in cluster analysis. Explain briefly. [3M] L2

   i) Identify the rules that describe specific patterns within the data. [2M] L1

   j) Discuss various kinds of patterns to be mined from web/server logs in web usage mining. [3M] L2

PART- B

(5 X 10 =50 Marks)

2. a) In real-world data, tuples with missing values for some attributes are a common occurrence. Describe various methods for handling this problem. [5M] L2

   b) State the various issues to consider during data integration. [5M] L2

OR

3. a) What are Major Issues in Data Mining? Describe in detail. [5M] L2

   b) Describe three challenges regarding data mining methodology and user interaction issues. [5M] L2

4. a) Write the Algorithm for Apriori to Find frequent itemset using an iterative level-wise approach based on candidate generation. [5M] L2

   b) Explain Graph pattern approach in data mining. Describe the approaches used in Graph pattern. [5M] L2

OR

5. Suppose we have the following dataset that has various transactions, and from this dataset, find the frequent itemset and generate the association rules using the Apriori algorithm: [10M] L4

TID	ITEMSETS
T1	A, B
T2	B, D
T3	B, C
T4	A, B, D
T5	A, C
T6	B, C
T7	A, C
T8	A, B, C, E
T9	A, B, C

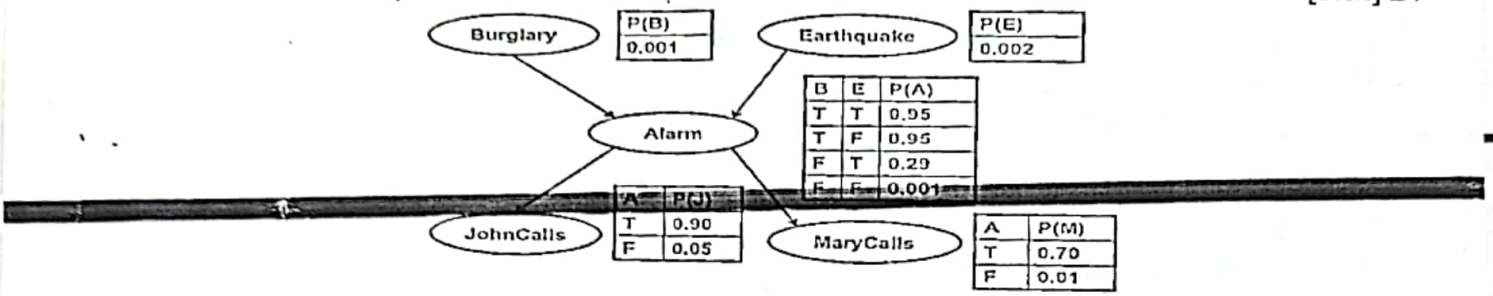
Given: Minimum Support= 2, Minimum Confidence= 50%

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6. How to apply the ID3 Algorithm to construct a decision tree? Discuss it with an example. [10M] L4

OR

7. What is the probability that an alarm has sounded, there was neither an earthquake nor a burglary, and both John and Mary called? [10M] L4



8. a) Describe the main difference in the approach of k-means and k-medoids algorithms with a neat diagram. [5M] L2

b) In a software project, the team is trying to identify the similarity of software defects identified during testing. They wanted to create 5 clusters of similar defects based on the text analytics of the defect descriptions. Once the 5 clusters of defects are identified, any new defect created is to be classified as one of the types identified through clustering. Explain this approach through a neat diagram. Assume 20 Defect data points which are clustered among 5 clusters and k-means algorithm was used. [5M] L3

OR

9. a) Appraise the importance of outlier detection and its application. Explain any one approach for outlier detection. [5M] L2

b) What is hierarchical clustering? Discuss the Single Linkage hierarchical clustering method with an example. [5M] L2

10. a) Explain the following. [5M] L2

(i) Spatial data mining (ii) Text mining.

b) Compare and contrast text mining with web content mining using lucid examples. [5M] L3

OR

11. a) What is web content mining? Discuss in detail. [5M] L2

b) Explain about unstructured text mining. [5M] L2

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year II Semester Supplementary Examinations, June -2024

DATA MINING

(INFORMATION TECHNOLOGY)

Time: 3 hours Max. Marks: 75

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (25 Marks)

1. a)	What is data preprocessing?	[2M] L1
b)	List out major issues in Data Preprocessing.	[3M] L1
c)	What is meant by association rule mining?	[2M] L1
d)	What is the need of confidence measure in association rule mining?	[3M] L2
e)	List the measures for selecting best split in decision tree construction	[2M] L1
f)	What is the significance of information gain?	[3M] L1
g)	List various clustering approaches.	[2M] L1
h)	What are primary features of density-based clustering?	[3M] L1
i)	What are stop words in text mining?	[2M] L1
j)	Give the taxonomy of web mining.	[3M] L2

PART- B (5 X 10 =50 Marks)

2. a)	Explain how to integrate data mining system with a data warehouse.	[5M] L2
b)	"Data preprocessing is necessary before data mining process". Justify your answer	[5M] L4

OR

3. a)	Discuss about challenging issues in Data Mining.	[5M] L2
b)	What is preprocessing? Explain about Data Transformation techniques	[5M] L2

4.	Explain Maximal Frequent Item Set and Closed Frequent Item Set.	[10M] L2

OR

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5. Explain the Apriori algorithm using the following transactional data assuming that the support count is 22%.

Illustrate with an example [10M] L3

TID	LISTOFITEMS
001	milk,dal,sugar,bread
002	Dal,sugar,wheat,jam
003	Milk,bread,curd,paneer
004	Wheat,paneer,dal,sugar
005	Milk,paneer,bread
006	Wheat,dal, paneer, bread

6. a) Discuss the significance of information gain in decision tree induction. [5M] L2

b) Explain Naïve-Bayesian classifier technique with an example. [5M] L2

OR

7. What is need of performing efficient implementation of Decision tree induction? Explain Entropy, Information gain and gini index with an example. [10M] L2

8. a) How density based method is used for clustering? [5M] L2

b) Illustrate K-mean algorithm with an example. [5M] L2

OR

9. a) Explain partitioning around mediods algorithm with an example. [5M] L2

b) Explain the key issues, strengths and weaknesses of hierarchical clustering algorithms. [5M] L2

10. a) Discuss the applications of web usage mining. [5M] L2

b) Explain web structure mining with a suitable algorithm. [5M] L2

OR

11. a) How to convert unstructured text into features in text mining? [5M] L3

b) Explain briefly about Web mining. [5M] L2

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech III Year II Semester Regular Examinations, July/August-2024

DATA MINING

(INFORMATION TECHNOLOGY)

Time: 3 hours | Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

PART- A (20 Marks)

1. a) Define data cleaning in the context of pre processing? [2M]L1
   b) Describe the purpose of the data transformation in data pre-processing stage? [2M]L2
   c) Explain concept of multi level Association rules? [2M]L2
   d) Explain the concept of confidence in Association mining? [2M]L2
   e) Explain difference between Classification and Prediction in data analysis? [2M]L2
   f) How Navy bayes classifier works? [2M]L1
   g) Describe basic principle of K-Means Algorithm. [2M]L2
   h) Interpret the meaning of dendrogram in Hierarchical clustering? [2M]L2
   i) List the 4 major components of trend analysis in time series data? [2M]L1
   j) Name 2 types of retrieval systems used for multimedia data? [2M]L1

PART- B (5 X 10 =50 Marks)

2. a) Explain about Data Mining Task Primitives. [5M]L2
   b) Give an Example of how you would assess the Interestingness of a pattern discovery in a Retail Dataset? [5M]L3

OR

3. Explain about Data Preprocessing. [10M]L2
4. With an example, explain the frequent itemset generation using Apriori Algorithm. [10M]L3

OR

5. a) Explain the types of Association Rules in Data Mining. [5M]L2
   b) Explain Market Basket Analysis ? [5M]L2

6. How to apply the ID3 Algorithm to construct a Decision Tree? Discuss with an Example. [10M]L4

OR

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7. Consider the given dataset, apply Navy based Algorithm and predict that if the fruit has the following properties, then which type of fruit is fruit={yellow, sweet, long} [10M]L4

Fruit	Yellow	Sweet	Long	Total
Orange	350	450	0	650
Banana	400	300	35	400
Others	50	100	400	150
Total	800	850	400	1200

8. Explain different methods of clustering? [10M]L2

OR

9. Explain the partitioning methods for clustering with an example? [10M]L3

10. a) How does the dynamic nature of World Wide Web effect the process of Web mining. [5M]L2

b) Apply the concept of the constraint-based mining to improve the efficiency of sequential pattern mining. [5M]L3

OR

11. Discuss various kinds of patterns to be mined from web/server logs in web usage mining. [10M]L2

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Supplementary Examinations, December-2025/January-2026

DATA MINING

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours **Max. Marks: 70**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A (20 Marks)

1. a) What is Classification in Data Mining? [2M]L1
   b) Define data warehouse. [2M]L1
   c) Why is pattern explosion a problem in frequent pattern mining? [2M]L1
   d) What is Graph pattern mining? [2M]L2
   e) What is entropy in decision tree learning? [2M]L2
   f) Define supervised learning. [2M]L2
   g) Name any two types of data used in cluster analysis. [2M]L1
   h) Differentiate between noise and outliers. [2M]L4
   i) How does spatial autocorrelation affect mining results? [2M]L3
   j) How does noise affect time-series mining? [2M]L5

PART- B (5 X 10 =50 Marks)

2. a) Design a data mining system integrated with a data warehouse. [5M]L5
   b) Explain the basic functionalities of data mining. [5M]L2

OR

3. a) What is data preprocessing? Explain its importance. [5M]L1
   b) Compare OLAP and data mining. [5M]L2

4. a) Describe frequent pattern mining in detail. [5M]L1
   b) Compare frequent, closed, and maximal pattern mining [5M]L4

OR

5. a) Evaluate support, confidence, and correlation measures in association rule mining. [5M]L2
   b) Describe the concept of constraint-based association mining. [5M]L1

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6. a) Describe the K-Nearest Neighbor classifier. [5M]L2

b) Describe decision tree induction with a simple example. [5M]L1

OR

7. a) Explain lazy learning and its advantages and disadvantages. [5M]L2

b) Evaluate the impact of rule pruning on classification performance. [5M]L6

8. a) Compare clustering and classification with examples [5M]L2

b) Describe the DBSCAN algorithm with its advantages and limitations. [5M]L1

OR

9. a) Compare agglomerative and divisive hierarchical clustering. [5M]L4

b) Explain density-based clustering methods. [5M]L2

10. a) Explain about Multimedia Data Mining in Detail. [5M]L1

b) Explain sequential pattern mining with a simple example. [5M]L2

OR

11. a) Analyze different methods for mining time-series data. [5M]L4

b) Explain the challenges involved in mining complex data types. [5M]L2

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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B.Tech IV Year I Semester Regular Examinations, December-2025/January-2026

DATA MINING

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours | Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory, which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub-questions.

PART- A

(20 Marks)

1. a)	What is multidimensional data?	[2M]L1
b)	Differentiate between discrete and continuous data.	[2M]L2
c)	Define support and confidence.	[2M]L2
d)	What is Apriori principle?	[2M]L1
e)	How does noise affect rule-based classifiers?	[2M]L2
f)	What is entropy in decision tree learning?	[2M]L1
g)	What is a dendrogram?	[2M]L1
h)	List any two applications of clustering.	[2M]L2
i)	What is web content mining?	[2M]L1
j)	Define time-series data.	[2M]L1

PART- B

(5 X 10 =50 Marks)

2. a)	Explain the concept of data mining task primitives.	[5M]L2
b)	Discuss data normalization techniques used in preprocessing.	[5M]L3
	OR	
3. a)	Discuss the need for integrating data mining with data warehouses.	[5M]L2
b)	Compare different classifications of data mining systems with real-time examples.	[5M]L4
4. a)	Explain association rule mining with a simple example.	[5M]L2
b)	Describe sequential pattern mining with an example.	[5M]L2
	OR	
5. a)	Describe frequent pattern mining in detail.	[5M]L2
b)	Evaluate support, confidence, and correlation measures in association rule mining	[5M]L4

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6. a) Discuss rule pruning techniques and their importance. [5M]L2

b) Compare eager learners and lazy learners with real-world examples. [5M]L4

OR

7. a) Explain Bayes' theorem and its role in classification. [5M]L2

b) Describe the K-Nearest Neighbor classifier. [5M]L2

8. a) Describe grid-based clustering methods. [5M]L2

b) Explain the K-means clustering algorithm with a suitable example. [5M]L2

OR

9. a) Explain hierarchical clustering with an example. [5M]L2

b) Explain the role of outlier analysis in clustering. [5M]L2

10. a) Describe the characteristics of time-series data. [5M]L2

b) Explain spatial data mining and its applications. [5M]L2

OR

11. a) Discuss different types of web mining. [5M]L2

b) Discuss text retrieval indexing techniques? [5M]L2

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Kmit

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Regular Examinations, January-2025

DATA MINING

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours **Max. Marks: 70**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A **(20 Marks)**

1. a)	What is nominal attribute? Give any two examples.	[2M] L1
b)	List three different binning methods in data cleaning.	[2M] L1
c)	Why frequent pattern growth algorithm is fast?	[2M] L2
d)	Write the formula to calculate confidence to generate association rules.	[2M] L1
e)	What is size ordering scheme in rule based classification?	[2M] L1
f)	Write the formula to calculate posteriori probability.	[2M] L1
g)	Describe dissimilarity matrix.	[2M] L2
h)	Define outliers in clustering.	[2M] L1
i)	Explain time series database.	[2M] L2
j)	What is the basic idea of the vector space model for text retrieval?	[2M] L1

12

PART- B **(5 X 10 =50 Marks)**

2. a)	Define data mining. Write a note on data mining architecture.	[5M] L1
b)	Write and explain major issues of data mining to derive various patterns.	[5M] L2

OR

3. a)	Illustrate data characterization and data discrimination handled in data mining.	[5M] L3
b)	Write a note on major tasks in data preprocessing.	[5M] L2
4. a)	Explain graph pattern mining and sequential pattern mining.	[4M] L2
b)	Apply Apriori algorithm to find frequent itemsets from the below dataset. Given minimum support count = 2.	[6M] L3

TID	Items
1	I1, I2, I5
2	I2, I4
3	I2, I3
4	I1, I2, I4
5	I1, I3
6	I2, I3
7	I1, I3
8	I1, I2, I3, I5
9	I1, I2, I3

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OR

5. a) A database of transactions in a book mart is as follows: Let min-sup = 25% [6M] L3

Trans_ID	Items
101	Book, Pen, Eraser
102	Pen, Pencil
103	Notebook, Book, Pen, Eraser
104	Book, Pen
105	Book, Notebook, Eraser

Using the abbreviations B for Book, P for Pen, E for Eraser, PN for Pencil and N for Notebook, find all frequent itemsets using FP tree algorithm. Construct FP Tree, conditional pattern base and conditional FP Tree.

b) Discuss the limitations of apriori algorithm. [4M] L3

6. a) Discuss the data preprocessing steps needed to proceed with classification and prediction. [5M] L2

b) Explain lazy learner and K-Nearest neighbor classifier. [5M] L2

OR

7. a) Write a note on case based reasoning. [5M] L2

b) Describe a procedure to derive classification by decision tree induction. [5M] L2

8. a) Demonstrate K-MEANS partitioning algorithm with any example. [5M] L3

b) Write a note on grid based clustering. [5M] L2

OR

9. a) Describe divisive clustering method with an example. [5M] L2

b) Explain density based methods of clustering. [5M] L2

10. a) Explain role of similarity search method in multimedia data mining. [5M] L2

b) Discuss various methods of spatial data mining in detail. [5M] L2

OR

11. a) Explain the concept of mining web page layout structure in World Wide Web. [5M] L2

b) Discuss about techniques for mining complex data types. [5M] L3

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KR21

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Supplementary Examinations, June-2025

DATA MINING

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours

Max. Marks: 70

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 20 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A

(20 Marks)

1. a) List two data mining functionalities. [2M] L1

b) What are task primitives in data mining? [2M] L2

c) Define support and confidence in association rule mining [2M] L1

d) What is a frequent item set? [2M] L2

e) What is decision tree pruning? [2M] L2

f) State Bayes' Theorem. [2M] L1

g) Mention two types of data used in clustering. [2M] L3

h) Name any two real-world applications of clustering. [2M] L2

i) What is the challenge in mining time-series data? [2M] L2

j) List two applications of spatial data mining. [2M] L3

PART- B

(5 X 10 =50 Marks)

2. a) Explain the classification of data mining systems based on techniques and applications. [5M] L2

b) Describe how data mining can be integrated with a data warehouse. [5M] L3

OR

3. Explain different data mining Functionalities with examples. [10M] L3

4. Explain the working of the Apriori algorithm with an example. [10M] L3

OR

5. Apply FP-growth to find frequent item sets in a dataset. [10M] L3

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6. a) Explain decision tree induction with an example [5M] L2

b) Compare classification and prediction with examples [5M] L3

OR

7. Apply Naïve Bayes to classify a small weather dataset. [10M] L4

8. Explain K-means clustering algorithm with an example. [10M] L2

OR

9. Explain different hierarchical methods with examples. [10M] L2

10. a) Explain the basic framework for mining data streams [5M] L2

b) Discuss sequence pattern mining in transactional databases. [5M] L2

OR

11. Evaluate the importance of mining time-series data in stock prediction. [10M] L3

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KR20

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Supplementary Examinations, May -2024

DATA MINING

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours

Max. Marks: 75

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b as sub questions.

PART- A

(25 Marks)

1. a)	What is data warehouse?	[2M]	L1
b)	List the methods of filling missing values.	[3M]	L1
c)	What is meant by association rule mining?	[2M]	L1
d)	What is the need of confidence measure in association rule mining?	[3M]	L1
e)	List the measures for selecting best split in decision tree construction.	[2M]	L1
f)	What is the significance of information gain?	[3M]	L1
g)	List various clustering approaches.	[2M]	L1
h)	What are primary features of density-based clustering?	[3M]	L1
i)	Write a short notes on Spatial Data mining?	[2M]	L1
j)	Give the taxonomy of web mining.	[3M]	L1

PART- B

(5 X 10 =50 Marks)

2. a)	Explain how to integrate data mining system with a data warehouse.	[5M]	L2
b)	"Data preprocessing is necessary before data mining process". Justify your answer	[5M]	L4

OR

3. a)	Discuss about challenging issues in Data Mining.	[5M]	L4
b)	What is preprocessing? Explain about Pre Processing techniques.	[5M]	L2
4. a)	Write a note on Maximal Frequent Item Set and Closed Frequent Item Set.	[5M]	L1
b)	Explain types of Association rules in Data Mining.	[5M]	L2

OR

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5. Explain market basket analysis and its relevance to association rule. Explain the Apriori algorithm using the following transactional data assuming that the support count is 22%. Illustrate with an example. [10M] L3

TID	LISTOFITEMS
001	milk,dal,sugar,bread
002	Dal,sugar,wheat,jam
003	Milk,bread,curd,paneer
004	Wheat,paneer,dal,sugar
005	Milk,paneer,bread
006	Wheat,dal, paneer, bread

6. a) Discuss the significance of information gain in decision tree induction. [5M] L4

b) Explain Naïve-Bayesian classifier technique with an example. [5M] L2

OR

7. What is need of performing efficient implementation of Decision tree induction? Explain Entropy, Information gain and gini index with an example. [10M] L2

8. a) How density based method is used for clustering? [5M] L2

b) Illustrate K-mean algorithm with an example. [5M] L2

OR

9. a) Explain partitioning around mediods algorithm with an example. [5M] L2

b) Explain the key issues, strengths and weaknesses of hierarchical clustering algorithms [5M] L2

10. a) Discuss the applications of web usage mining. [5M] L4

b) Explain web structure mining with a suitable algorithm. [5M] L2

OR

11. a) How to convert unstructured text into features in text mining? [5M] L2

b) Explain briefly about Web mining. [5M] L2

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KR20

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTE)

Approved by AICTE, Affiliated to JNTUH, Hyderabad-500 029

B.Tech IV Year I Semester Regular Examinations, December -2023

DATA MINING

(COMPUTER SCIENCE AND ENGINEERING)

Time: 3 hours

Max. Marks: 75

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1. a) What is data integration and data reduction? [2M], L2
   b) What is noisy data and write any two ways to handle them. [3M], L2
   c) Define Support and Confidence in Association rule. [2M], L2
   d) Write about Multilevel association rules and Multidimensional association rules. [3M], L2
   e) Write any two issues regarding Classification & Prediction. [2M], L2
   f) Write any three differences between eager and lazy learner classification methods. [3M], L2
   g) Define cluster analysis and outlier analysis. [2M], L2
   h) Differentiate agglomerative clustering from divisive clustering. [3M], L2
   i) What do you mean by spatial data mining? [2M], L2
   j) What is a multimedia database? What kinds of associations can be mined in multimedia data? [3M], L2

PART- B

(5 X 10 =50 Marks)

2. a) Categorize and explain the different kinds of data that can be mined. [5M], L4
   b) Explain about all the issues in Data Mining research related to Diversity of Database types and Data mining and society. [5M], L3

OR

3. a) List out major data mining functionalities. Explain any two of them. [5M], L3
   b) Discuss about the major tasks in Data Preprocessing. [5M], L5

4. a) Examine how different types of correlation measures are used in Data mining framework for association rules. [5M], L4
   b) Discuss in detail about Graph Pattern Mining and Sequential Pattern Mining. [5M], L5

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OR

5. Some transactions in Big Bazaar are as follows:

TID	List of items
T100	Colour TV, DVD, Speakers
T200	DVD, Laptop
T300	DVD, Desktop
T400	Colour TV, DVD, Laptop
T500	Colour TV, DVD, Desktop

The minimum support = 30%. Using C for Colour TV, D for DVD, DT for Desktop, L for Laptop and S for Speakers, draw the FP Tree and corresponding conditional pattern bases along with conditional FP Tree. [10M], L3

6. a) Write about classification with an example using decision tree induction approach. [5M], L3
   b) What are Bayesian classifiers? Discuss the working of naïve Bayesian classifier. [5M], L3

OR

7. a) Explain the steps involved in decision tree classification with a suitable example. [5M], L3
   b) Compare the advantages and disadvantages of eager classification (e.g., decision tree, Bayesian, neural network) versus lazy classification. [5M], L4

8. a) Explain K-mediods clustering with an example. [5M], L3
   b) Explain with an example how DBSCAN quantify the neighborhood of an object. [5M], L3

OR

9. a) Explain in detail about Grid-based clustering methods. [5M], L6
   b) What are outliers? Explain various outlier detection methods. [5M], L2

10. a) Analyze about spatial data mining. [5M], L4
    b) Explain in detail about the Text Data Analysis and Information Retrieval. [5M], L3

OR

11. a) Explain about mining of the World Wide Web. [5M], L3
    b) The Web also poses great challenges for effective resource and knowledge discovery. Elaborate. [5M], L5

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